

Python Code

```
1  from sqlalchemy import (
2      create_engine, Column, Integer, String, Date, Numeric,
3      ForeignKey, UniqueConstraint
4  )
5  from sqlalchemy.orm import declarative_base, relationship
6
7  engine = create_engine(
8      "mysql+mysqlconnector://root:Sugii123%40@localhost/college_db_orm",
9      echo=True
10 )
11
12  Base = declarative_base()
13
14
15  class Department(Base):
16      __tablename__ = "departments"
17
18      dept_id = Column(Integer, primary_key=True, autoincrement=True)
19      dept_name = Column(String(100), nullable=False, unique=True)
20      head_of_dept = Column(String(100), nullable=False)
21      location = Column(String(100))
22      budget = Column(Numeric(12, 2))
23
24      students = relationship("Student", back_populates="department")
25      courses = relationship("Course", back_populates="department")
26      professors = relationship("Professor", back_populates="department")
27
28
29  class Student(Base):
30      __tablename__ = "students"
31
32      student_id = Column(Integer, primary_key=True, autoincrement=True)
33      first_name = Column(String(50), nullable=False)
34      last_name = Column(String(50), nullable=False)
35      email = Column(String(100), nullable=False, unique=True)
36      dob = Column(Date, nullable=False)
37      dept_id = Column(Integer, ForeignKey("departments.dept_id"), nullable=False)
38      enrollment_year = Column(Integer)
39
40      department = relationship("Department", back_populates="students")
41      enrollments = relationship("Enrollment", back_populates="student")
42
43
44  class Course(Base):
45      __tablename__ = "courses"
46
47      course_id = Column(Integer, primary_key=True, autoincrement=True)
48      course_name = Column(String(150), nullable=False)
49      credits = Column(Integer, nullable=False)
50      dept_id = Column(Integer, ForeignKey("departments.dept_id"), nullable=False)
```

```
51     max_seats = Column(Integer, default=60)
52
53     department = relationship("Department", back_populates="courses")
54     enrollments = relationship("Enrollment", back_populates="course")
55
56
57 class Professor(Base):
58     __tablename__ = "professors"
59
60     professor_id = Column(Integer, primary_key=True, autoincrement=True)
61     first_name = Column(String(50), nullable=False)
62     last_name = Column(String(50), nullable=False)
63     email = Column(String(100), nullable=False, unique=True)
64     hire_date = Column(Date)
65     dept_id = Column(Integer, ForeignKey("departments.dept_id"), nullable=False)
66     salary = Column(Numeric(10, 2))
67
68     department = relationship("Department", back_populates="professors")
69
70
71 class Enrollment(Base):
72     __tablename__ = "enrollments"
73
74     __table_args__ = (
75         UniqueConstraint("student_id", "course_id"),
76     )
77
78     enrollment_id = Column(Integer, primary_key=True, autoincrement=True)
79     student_id = Column(Integer, ForeignKey("students.student_id"), nullable=False)
80     course_id = Column(Integer, ForeignKey("courses.course_id"), nullable=False)
81     enrolled_on = Column(Date)
82     grade = Column(String(1))
83
84     student = relationship("Student", back_populates="enrollments")
85     course = relationship("Course", back_populates="enrollments")
86
87
88 if __name__ == "__main__":
89     Base.metadata.create_all(engine)
90     print("All 5 tables created successfully!")
```

Python Code

```
1  # =====
2  # Query count comparison (Step 90)
3  #
4  # Lazy Loading:
5  # 1 query for enrollments +
6  # N queries for students +
7  # N queries for courses
8  #
9  # Eager Loading (joinedload):
10 # Only 1 SQL query with JOINS.
11 # =====
12
13 from datetime import date
14 from sqlalchemy import create_engine
15 from sqlalchemy.orm import sessionmaker, joinedload
16
17 from models import (
18     Department,
19     Student,
20     Course,
21     Enrollment,
22     Professor
23 )
24
25 # -----
26 # Database Connection
27 # -----
28
29 engine = create_engine(
30     "mysql+mysqlconnector://root:Sugii123%40@localhost/college_db_orm",
31     echo=True
32 )
33
34 Session = sessionmaker(bind=engine)
35 session = Session()
36
37 # =====
38 # STEP 81
39 # INSERT Departments
40 # =====
41
42 departments = [
43
44     Department(
45         dept_name="Computer Science",
46         head_of_dept="Dr. Alan Turing",
47         location="Block A",
48         budget=750000.00
49     ),
50
```

```
51     Department(  
52         dept_name="Mathematics",  
53         head_of_dept="Dr. Emmy Noether",  
54         location="Block B",  
55         budget=620000.00  
56     ),  
57  
58     Department(  
59         dept_name="Electronics",  
60         head_of_dept="Dr. Nikola Tesla",  
61         location="Block C",  
62         budget=580000.00  
63     )  
64  
65 ]  
66  
67 session.add_all(departments)  
68 session.commit()  
69  
70 # -----  
71  
72 cs_dept = session.query(Department).filter_by(  
73     dept_name="Computer Science"  
74 ).first()  
75  
76 math_dept = session.query(Department).filter_by(  
77     dept_name="Mathematics"  
78 ).first()  
79  
80 ec_dept = session.query(Department).filter_by(  
81     dept_name="Electronics"  
82 ).first()  
83  
84 # =====  
85 # STEP 81  
86 # INSERT Students  
87 # =====  
88  
89 students = [  
90  
91     Student(  
92         first_name="Alice",  
93         last_name="Johnson",  
94         email="alice.johnson@college.edu",  
95         dob=date(2001,3,15),  
96         dept_id=cs_dept.dept_id,  
97         enrollment_year=2021  
98     ),  
99  
100    Student(  
101        first_name="Bob",  
102        last_name="Smith",  
103        email="bob.smith@college.edu",  
104        dob=date(2000,7,22),
```

```
105         dept_id=cs_dept.dept_id,
106         enrollment_year=2021
107     ),
108
109     Student(
110         first_name="Carol",
111         last_name="Williams",
112         email="carol.williams@college.edu",
113         dob=date(2002,1,10),
114         dept_id=math_dept.dept_id,
115         enrollment_year=2022
116     ),
117
118     Student(
119         first_name="David",
120         last_name="Brown",
121         email="david.brown@college.edu",
122         dob=date(2001,11,5),
123         dept_id=math_dept.dept_id,
124         enrollment_year=2022
125     ),
126
127     Student(
128         first_name="Eva",
129         last_name="Davis",
130         email="eva.davis@college.edu",
131         dob=date(2002,6,30),
132         dept_id=ec_dept.dept_id,
133         enrollment_year=2022
134     )
135 ]
136
137 session.add_all(students)
138 session.commit()
139
140
141 # =====
142 # STEP 82
143 # INSERT Courses
144 # =====
145
146 courses = [
147
148     Course(
149         course_name="CS101 - Intro to Programming",
150         credits=4,
151         dept_id=cs_dept.dept_id
152     ),
153
154     Course(
155         course_name="CS201 - Data Structures",
156         credits=4,
157         dept_id=cs_dept.dept_id
158     ),
```

```
159
160     Course(
161         course_name="MA101 - Calculus I",
162         credits=3,
163         dept_id=math_dept.dept_id
164     )
165
166 ]
167
168 session.add_all(courses)
169 session.commit()
170
171 # -----
172
173 alice = session.query(Student).filter_by(
174     email="alice.johnson@college.edu"
175 ).first()
176
177 bob = session.query(Student).filter_by(
178     email="bob.smith@college.edu"
179 ).first()
180
181 carol = session.query(Student).filter_by(
182     email="carol.williams@college.edu"
183 ).first()
184
185 cs101 = session.query(Course).filter_by(
186     course_name="CS101 - Intro to Programming"
187 ).first()
188
189 cs201 = session.query(Course).filter_by(
190     course_name="CS201 - Data Structures"
191 ).first()
192
193 ma101 = session.query(Course).filter_by(
194     course_name="MA101 - Calculus I"
195 ).first()
196
197 # =====
198 # STEP 82
199 # INSERT Enrollments
200 # =====
201
202 enrollments = [
203
204     Enrollment(
205         student_id=alice.student_id,
206         course_id=cs101.course_id,
207         enrolled_on=date(2021,8,1),
208         grade="A"
209     ),
210
211     Enrollment(
212         student_id=alice.student_id,
```

```
213         course_id=cs201.course_id,
214         enrolled_on=date(2021,8,1),
215         grade="B"
216     ),
217
218     Enrollment(
219         student_id=bob.student_id,
220         course_id=cs101.course_id,
221         enrolled_on=date(2021,8,1),
222         grade="B"
223     ),
224
225     Enrollment(
226         student_id=carol.student_id,
227         course_id=ma101.course_id,
228         enrolled_on=date(2022,8,1),
229         grade="A"
230     )
231
232 ]
233
234 session.add_all(enrollments)
235 session.commit()
236
237 # =====
238 # STEP 83
239 # READ
240 # =====
241
242 print("\nStudents in Computer Science\n")
243
244 cs_students = (
245
246     session.query(Student)
247     .join(Department)
248     .filter(Department.dept_name=="Computer Science")
249     .all()
250
251 )
252
253 for s in cs_students:
254     print(s.first_name, s.last_name)
255
256 # =====
257 # STEP 84
258 # Lazy Loading
259 # =====
260
261 print("\nLazy Loading (watch SQL queries)\n")
262
263 all_enrollments = session.query(Enrollment).all()
264
265 for e in all_enrollments:
266     print(
```

```
267         e.student.first_name,
268         e.student.last_name,
269         "->",
270         e.course.course_name
271     )
272
273     # =====
274     # STEP 85
275     # UPDATE
276     # =====
277
278     student = session.query(Student).filter_by(
279         email="bob.smith@college.edu"
280     ).first()
281
282     student.enrollment_year = 2023
283
284     session.commit()
285
286     print("\nUpdated Bob's enrollment year.")
287
288     # =====
289     # STEP 86
290     # DELETE
291     # =====
292
293     enrollment = (
294         session.query(Enrollment)
295         .filter_by(
296             student_id=carol.student_id,
297             course_id=ma101.course_id
298         )
299         .first()
300     )
301
302     session.delete(enrollment)
303
304     session.commit()
305
306     print("Deleted Carol's enrollment.")
307
308     print("Remaining Enrollments:",
309           session.query(Enrollment).count())
310
311     # =====
312     # STEP 88
313     # joinedload()
314     # =====
315
316     print("\nEager Loading using joinedload()\n")
317
318     records = (
```

```
321
322     session.query(Enrollment)
323
324     .options(
325
326         joinedload(Enrollment.student),
327
328         joinedload(Enrollment.course)
329
330     )
331
332     .all()
333
334 )
335
336 for e in records:
337
338     print(
339
340         e.student.first_name,
341
342         e.student.last_name,
343
344         "->",
345
346         e.course.course_name
347
348     )
349
350 session.close()
```

```
PS C:\Users\sugxn\OneDrive\Documents\Projects\CTS\Module 1\hands_on_6> python models.py
2026-06-27 13:03:04,868 INFO sqlalchemy.engine.Engine SELECT DATABASE()
2026-06-27 13:03:04,869 INFO sqlalchemy.engine.Engine [raw sql] {}
2026-06-27 13:03:04,870 INFO sqlalchemy.engine.Engine SELECT @@sql_mode
2026-06-27 13:03:04,870 INFO sqlalchemy.engine.Engine [raw sql] {}
2026-06-27 13:03:04,871 INFO sqlalchemy.engine.Engine SELECT @@lower_case_table_names
2026-06-27 13:03:04,871 INFO sqlalchemy.engine.Engine [raw sql] {}
2026-06-27 13:03:04,872 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-06-27 13:03:04,872 INFO sqlalchemy.engine.Engine DESCRIBE `college_db_orm`.`departments`
2026-06-27 13:03:04,873 INFO sqlalchemy.engine.Engine [raw sql] {}
2026-06-27 13:03:04,875 INFO sqlalchemy.engine.Engine DESCRIBE `college_db_orm`.`students`
2026-06-27 13:03:04,877 INFO sqlalchemy.engine.Engine [raw sql] {}
2026-06-27 13:03:04,878 INFO sqlalchemy.engine.Engine DESCRIBE `college_db_orm`.`courses`
2026-06-27 13:03:04,879 INFO sqlalchemy.engine.Engine [raw sql] {}
2026-06-27 13:03:04,880 INFO sqlalchemy.engine.Engine DESCRIBE `college_db_orm`.`professors`
2026-06-27 13:03:04,880 INFO sqlalchemy.engine.Engine [raw sql] {}
2026-06-27 13:03:04,884 INFO sqlalchemy.engine.Engine DESCRIBE `college_db_orm`.`enrollments`
2026-06-27 13:03:04,888 INFO sqlalchemy.engine.Engine [raw sql] {}
2026-06-27 13:03:04,896 INFO sqlalchemy.engine.Engine
CREATE TABLE departments (
    dept_id INTEGER NOT NULL AUTO_INCREMENT,
    dept_name VARCHAR(100) NOT NULL,
    head_of_dept VARCHAR(100) NOT NULL,
    location VARCHAR(100),
    budget NUMERIC(12, 2),
    PRIMARY KEY (dept_id),
    UNIQUE (dept_name)
)

2026-06-27 13:03:04,898 INFO sqlalchemy.engine.Engine [no key 0.00471s] {}
2026-06-27 13:03:04,937 INFO sqlalchemy.engine.Engine
CREATE TABLE students (
    student_id INTEGER NOT NULL AUTO_INCREMENT,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL,
    dob DATE NOT NULL,
    dept_id INTEGER NOT NULL,
    enrollment_year INTEGER,
    PRIMARY KEY (student_id),
    UNIQUE (email),
    FOREIGN KEY(dept_id) REFERENCES departments (dept_id)
)

2026-06-27 13:03:04,937 INFO sqlalchemy.engine.Engine [no key 0.00052s] {}
2026-06-27 13:03:04,966 INFO sqlalchemy.engine.Engine
CREATE TABLE courses (
    course_id INTEGER NOT NULL AUTO_INCREMENT,
    course_name VARCHAR(150) NOT NULL,
    credits INTEGER NOT NULL,
```